CISCO INTERNSHIP · SUMMARY REPORT

FaultLine

AI-Assisted Network Fault Diagnosis & Remediation System

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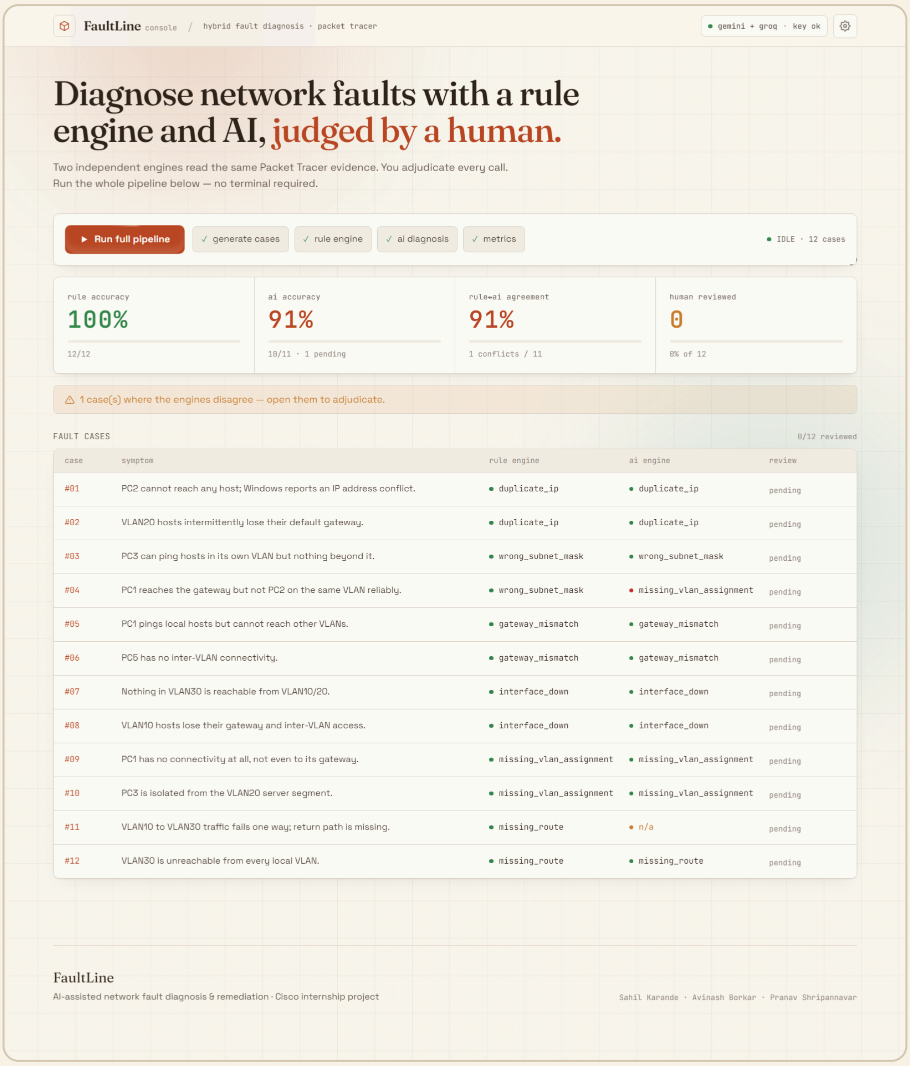
Manual network troubleshooting is slow and inconsistent. FaultLine is a hybrid diagnostic pipeline that reads real Cisco Packet Tracer command output and identifies the injected fault two independent ways — a deterministic **Python/TypeScript rule engine** and an **LLM reasoning module** — then places a **human in the loop** to adjudicate every AI recommendation before it is applied and verified.

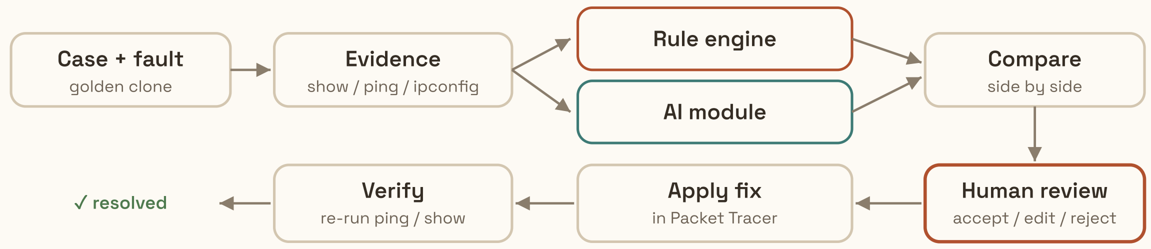
100% Rule-engine accuracy (12/12 cases)	**~91%** AI accuracy (Gemini + Groq)	**12** Fault cases · 6 fault types	**2** Independent engines, compared



FaultLine console — rule vs. AI per case, live metrics, and the human-review queue. Here the rule engine is 12/12 while the AI is 10/11; case #04 is a genuine rule↔AI disagreement flagged for human adjudication.

01 How it works

Every case is one clone of a "golden" topology (2 routers, a switch with VLANs 10/20/30, several PCs) with **exactly one injected fault**. The same collected evidence — `show ip interface brief`, `show ip route`, `show vlan brief`, `ipconfig`, `ping` — is fed to both engines in parallel, then compared. The novelty is the comparison + human-review stage: nothing the AI suggests is treated as final until a person accepts, edits, or rejects it.



02 Fault case bank

Twelve cases — two per fault type — each with a reported symptom and ground-truth label:

FAULT TYPE	WHAT IS INJECTED	EXAMPLE SYMPTOM
`duplicate_ip`	Two hosts share one IP	PC reports an IP address conflict
`wrong_subnet_mask`	Host mask ≠ segment mask	Local reachable, remote not
`gateway_mismatch`	Default gateway points nowhere	Local OK, no inter-VLAN
`interface_down`	Router/trunk port shut	A whole segment unreachable
`missing_vlan_assignment`	Access port left in VLAN 1	Host fully isolated
`missing_route`	Static route removed	One-way / remote unreachable

03 Results & the human-in-the-loop

The rule engine is deterministic and scores **12/12** — it parses the evidence and diffs it against the golden baseline. The AI (Gemini + Groq) typically lands **10–12 / 12**; when it disagrees with the rule engine or fails to answer, the case is surfaced for a human to adjudicate. That disagreement is the point: the AI is advisory, never autonomous.

METRIC	RESULT	NOTES
Rule-engine accuracy	`12 / 12 = 100%`	Each a single high-confidence finding
AI accuracy	`~10-12 / 12`	Scored over cases actually diagnosed
Rule ↔ AI agreement	`~90-100%`	Disagreements routed to human review
Human decisions	`accept / edit / reject`	Every decision logged (Postgres)
Time to diagnosis	`< 1 s rule · 2-6 s AI`	vs. 5–15 min manual

04 System & deployment

Frontend

- Vue 3 + Vite, Inspira UI motion components
- Editorial theme (Hallmark), parallax background
- Runs the whole pipeline in-browser — no terminal

`Vue 3` `Vite` `Tailwind v4` `Fraunces` `Vercel`

Backend

- Supabase Edge Function (Deno) — rule engine + AI + metrics
- Reviews & diagnoses persisted in Postgres
- Two LLM providers with per-case fallback

`Supabase` `Deno` `Postgres` `Gemini` `Groq`

Also on mobile

The console is fully responsive — the same run-the-pipeline, review-every-call flow, stacked for a phone.

Live · cisco-project-a1jk.vercel.app
Repo · github.com/sahilkarande0918-cmd/CISCO-Project

Responsive mobile layout.

05 What makes it a real project

Novelty

Two independent engines on identical evidence, with a logged human decision on every AI call — governance you can point to, not just a model output.

Engineering

Deterministic parser (regex → JSON, ported Python→TypeScript, verified 12/12), resilient two-provider AI with timeouts + fallback, and live per-case progress.

06 Per-case pipeline

1. Collect evidence for the case (real Packet Tracer output).
2. Rule engine + AI diagnose in parallel; results compared side by side.
3. Human accepts / edits / rejects the AI recommendation — decision logged.
4. Apply the accepted fix in Packet Tracer.
5. Re-run `ping` / `show` to confirm the symptom is gone → **verified**.

Team

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Cisco internship · AI-Assisted Network Fault Diagnosis & Remediation System

FaultLine turns manual, inconsistent troubleshooting into a fast, explainable, human-governed pipeline — deterministic where it can be, AI-assisted where it helps, and always adjudicated by a person before a fix lands.